

Frontend Development & Backend Integration

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
Instructor	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
Section	7C
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GitHub Repository Link	cse4104-7c-t05-ai-academic-mentor/documentation at main · Priyo-1158/ cse4104-7c-t05-ai-academic-mentor · GitHub

Team Information

Role	Name	Student ID	Contribution
Team Leader	Shadman Sadequeen Priyo	11230121158	Project coordination, GitHub management, integration testing
Frontend Developer	Musabbir Hossain Chayon	11230121168	API integration, Report, Ulpages, Netlify deployment, Next.js
Backend Developer	SK. Nadirul Haque Nadir	11230121155	Node.js, Express, routing, Express routes, MongoDB models, JWT authentication
AI Integration Lead	Shafin Kabir	11230121175	Gemini API, HuggingFace, AI routes

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1. Project Overview

Project Description

AI Academic Mentor is a full-stack, AI-assisted study platform providing authenticated users with an AI chatbot, quiz generation, note summarization, an adaptive study planner, progress tracking, and an administrative dashboard for platform oversight. The system follows a decoupled architecture: a React single-page application communicates with an independent Express/MongoDB REST API, with all AI provider keys held strictly server-side.

2. Frontend Technology Stack

Frontend

- **React 19** — component-based SPA architecture, function components + hooks throughout
- **Vite** — dev server and production bundler
- **React Router v7** — client-side routing with protected/admin route guards
- **Axios** — HTTP client with a centralized instance, JWT auth interceptor, and normalized error handling

Backend (integrated against)

- **Node.js + Express** — REST API server
- **MongoDB Atlas + Mongoose** — data persistence for Users, Notes, QuizResults, StudyPlans
- **JWT (jsonwebtoken) + bcryptjs** — stateless authentication and password hashing
- **Helmet + express-rate-limit** — security headers and brute-force mitigation

AI Layer

- **Google Gemini API** (primary) with HuggingFace Inference as fallback — proxied entirely through the backend; the frontend never holds an AI provider key

3. Completed Pages / Screens

All pages required by the Week 07 brief are implemented and routed, exceeding the minimum required screen set.

Page	Route	Status	Notes
Landing / Home	/	Yes	Marketing/entry page, unauthenticated
Login	/login	Yes	JWT-based sign-in
Registration	/register	Yes	New account creation
Dashboard	/dashboard	Yes	Protected — authenticated landing view
AI Chatbot	/chatbot	Yes	Protected — AI interaction page
Quiz	/quiz	Yes	Protected — AI-generated quiz flow
Summarizer	/summarizer	Yes	Protected — note summarization
Study Planner	/planner	Yes	Protected — AI-generated study plan
Progress Tracking	/progress	Yes	Protected — learning progress view
Notifications	/notifications	Yes	Protected — user notifications
Settings	/settings	Yes	Protected — account/profile settings
Admin Panel	/admin	Partial	Admin-only — core stats/users functional; Announcements & Bulk Wipe are disabled stubs (see §8)

4. Backend API Integration

The frontend communicates with the backend exclusively through a centralized Axios instance (services/api.js) that automatically attaches the JWT bearer token to every request and normalizes error responses to a consistent shape for UI consumption.

Module	Endpoint(s)	Consumed By	Status
Auth	POST /api/auth/register, /login	Registration & Login pages	Live
AI Proxy	POST /api/ai/chat, /quiz, /summarize, /plan	Chatbot, Quiz, Summarizer, Planner	Live plumbing / degraded output (\$8)
Quiz Results	GET/POST /api/quiz	Quiz, Progress	Live
Study Plans	GET/POST /api/plan	Planner, Dashboard	Live
Summaries	GET/POST /api/summary	Summarizer	Live
Progress	GET /api/progress	Progress, Dashboard	Live
Admin	GET/PUT/DELETE /api/admin/*	Admin Panel	Live (core) / stubbed (Announcements, Wipe)

Design note: the Gemini API key is never exposed to the client — all AI calls are proxied through the backend, per the architectural fix applied earlier in the project.

5. Authentication Flow

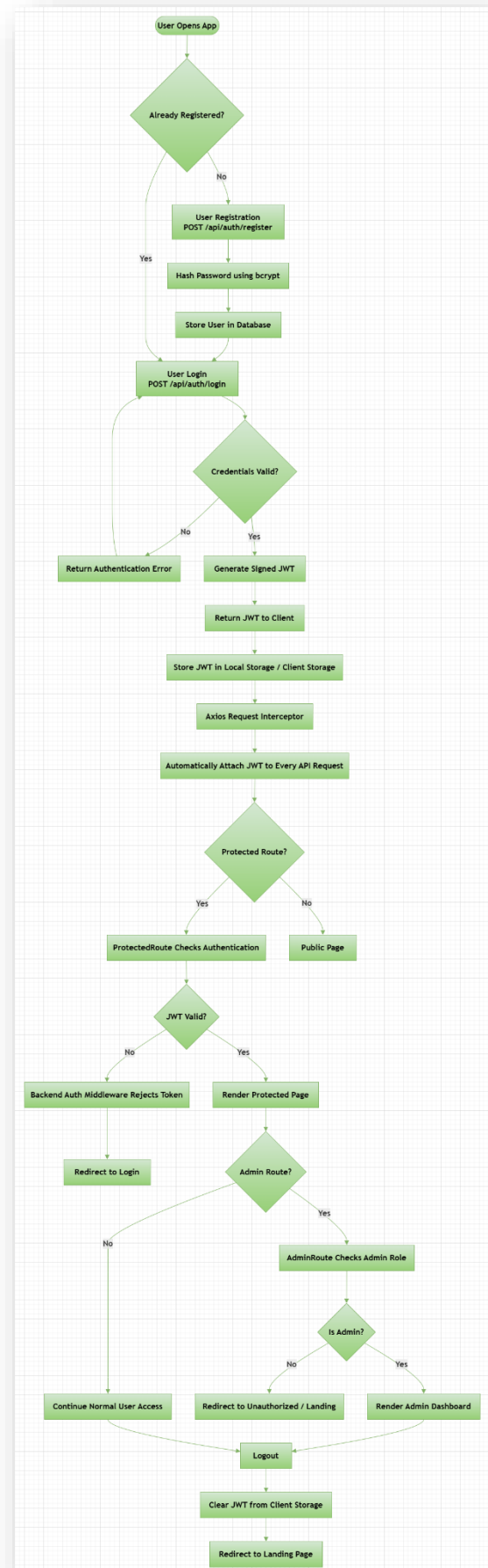
Implemented

- ✓ User Registration — POST /api/auth/register, bcrypt password hashing
- ✓ User Login — POST /api/auth/login, returns signed JWT
- ✓ Session persistence — JWT stored client-side, attached via Axios request interceptor
- ✓ Protected Routes — <ProtectedRoute> wraps all authenticated pages
- ✓ Admin-only Routes — <AdminRoute> gate on the /admin path
- ✓ Unauthorized Access Handling — unauthenticated/expired requests redirect via route guards
- ✓ Logout — clears stored session and redirects to Landing

Flow Summary

Register → credentials hashed and persisted → Login → backend issues JWT → token persisted client-side and attached automatically to every subsequent API call → protected pages check auth state before rendering → expired/invalid tokens are rejected by backend middleware (middleware/auth.js) and the user is routed back to Login.

Frontend Authentication Flow Diagram :



6. Current Development Progress

Completed this cycle

- Full page set implemented and routed (12 screens, exceeding the assignment's minimum requirement)
- Authentication flow fully wired end-to-end: register, login, JWT persistence, protected + admin routing, logout
- Centralized API layer with auth interceptor and normalized error handling across all pages
- AI proxy endpoints connected for Chatbot, Quiz, Summarizer, and Planner (backend plumbing complete)
- CRUD integration for quiz results, study plans, and summaries against MongoDB Atlas
- Admin dashboard core (stats, user list, user detail/activity, update/delete) functional
- Security hardening carried over from backend phase: Helmet headers, rate limiting, server-side key isolation

Outstanding before final submission

- Replace placeholder production API base URL with the live deployed backend URL
- Provision valid Gemini / HuggingFace API keys so AI features return live model output instead of canned fallback text
- Decide and document the fate of Admin Announcements and Bulk Clear/Wipe (currently disabled stubs)
- Capture and attach required UI screenshots (desktop + mobile) across all major screens
- Cross-device responsive verification pass (desktop / tablet / mobile)

7. Form Validation

Required: required fields, email format, password validation, invalid-input handling, user-friendly error messages.

Requirement	Implementation	Status
Required fields	required attribute on all auth/functional forms	DONE
Email format validation	type="email" on all email inputs	DONE
Password validation	minlength=6 + confirm-password match check	DONE
Invalid input handling	Server re-validates everything client already checked (Mongoose schemas)	DONE
User-friendly error messages	Styled .alert-error/.alert-success boxes replace raw alert() popups	DONE

8. User Experience Improvements

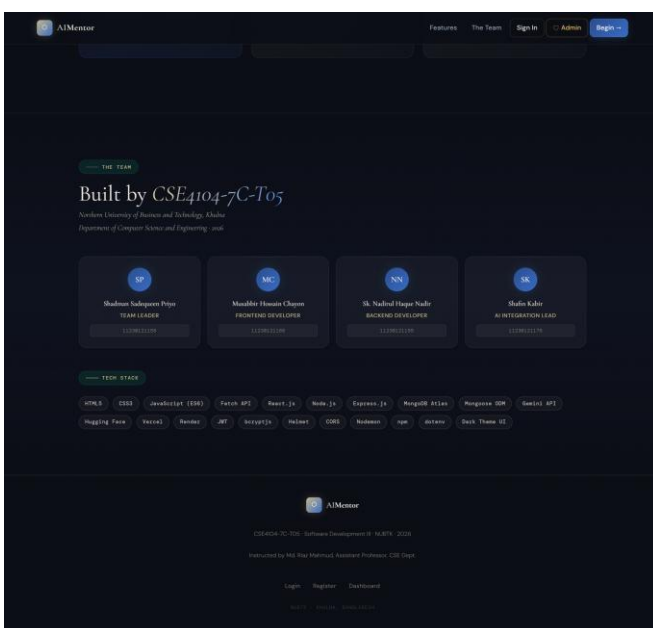
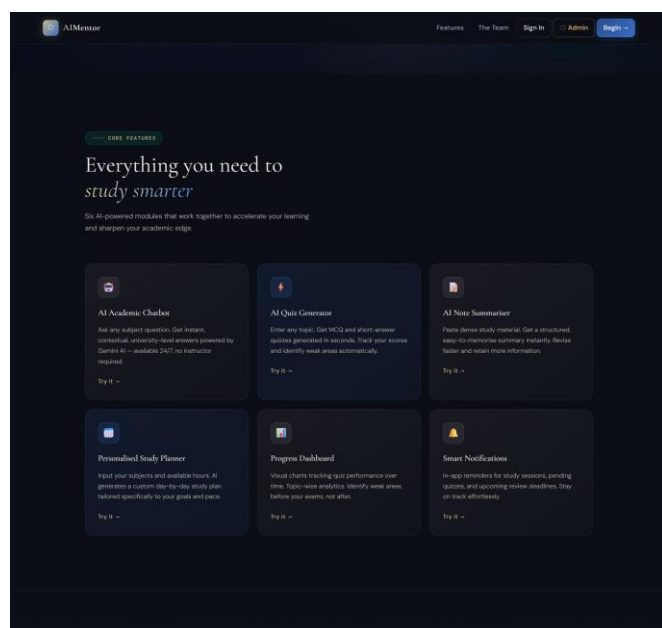
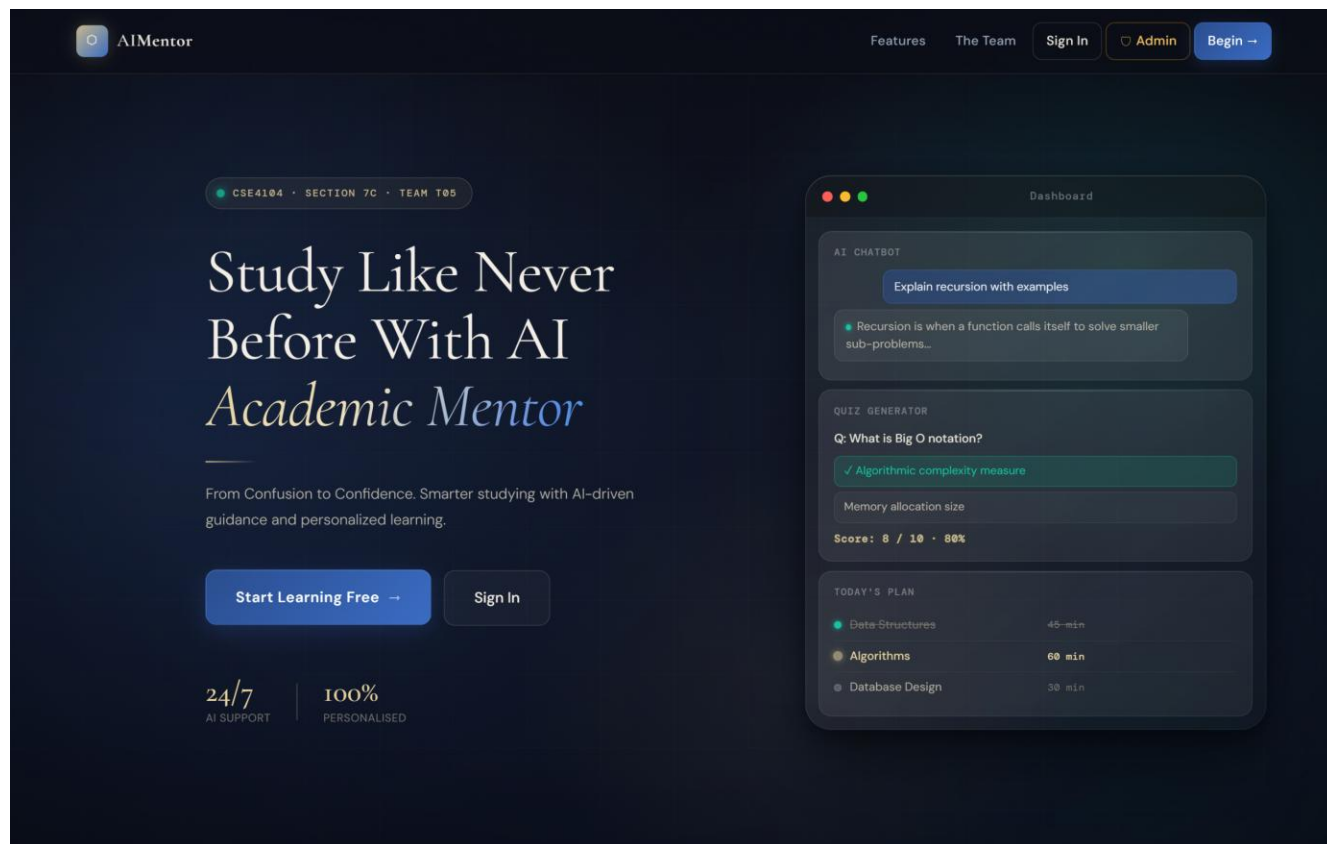
Suggested: loading indicators, success messages, error messages, empty-state messages, confirmation dialogs, better navigation.

UX Element	Present?	Status
Loading indicators	Spinners on submit buttons	DONE
Success messages	Styled .alert-success banners	DONE
Error messages	Styled .alert-error banners	DONE
Empty-state messages	Seen on Dashboard, Quiz History, Notifications	DONE
Confirmation dialogs	None found anywhere in the codebase — e.g. admin delete-user has no confirm step	PARTIAL
Better navigation	Persistent sidebar, active-page highlight, role-based nav item hiding	DONE

9. Responsive Design(UI Screenshots)

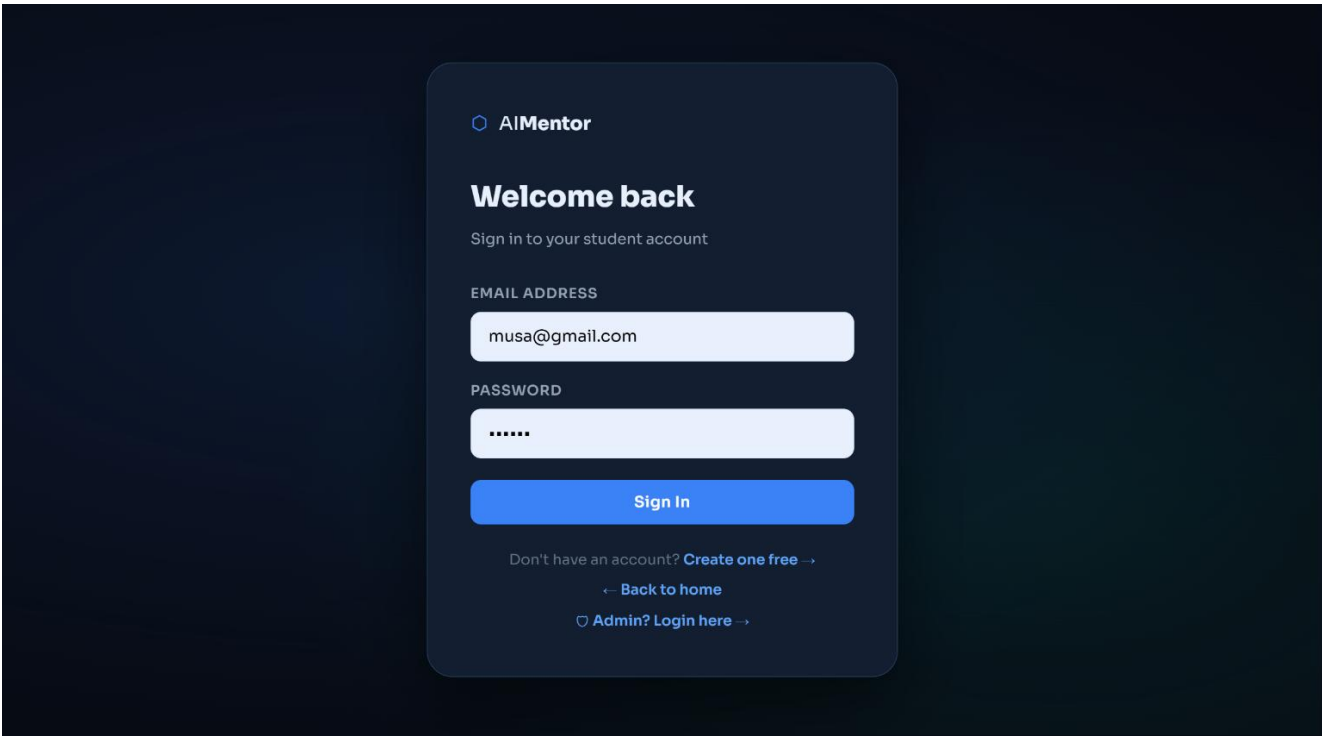
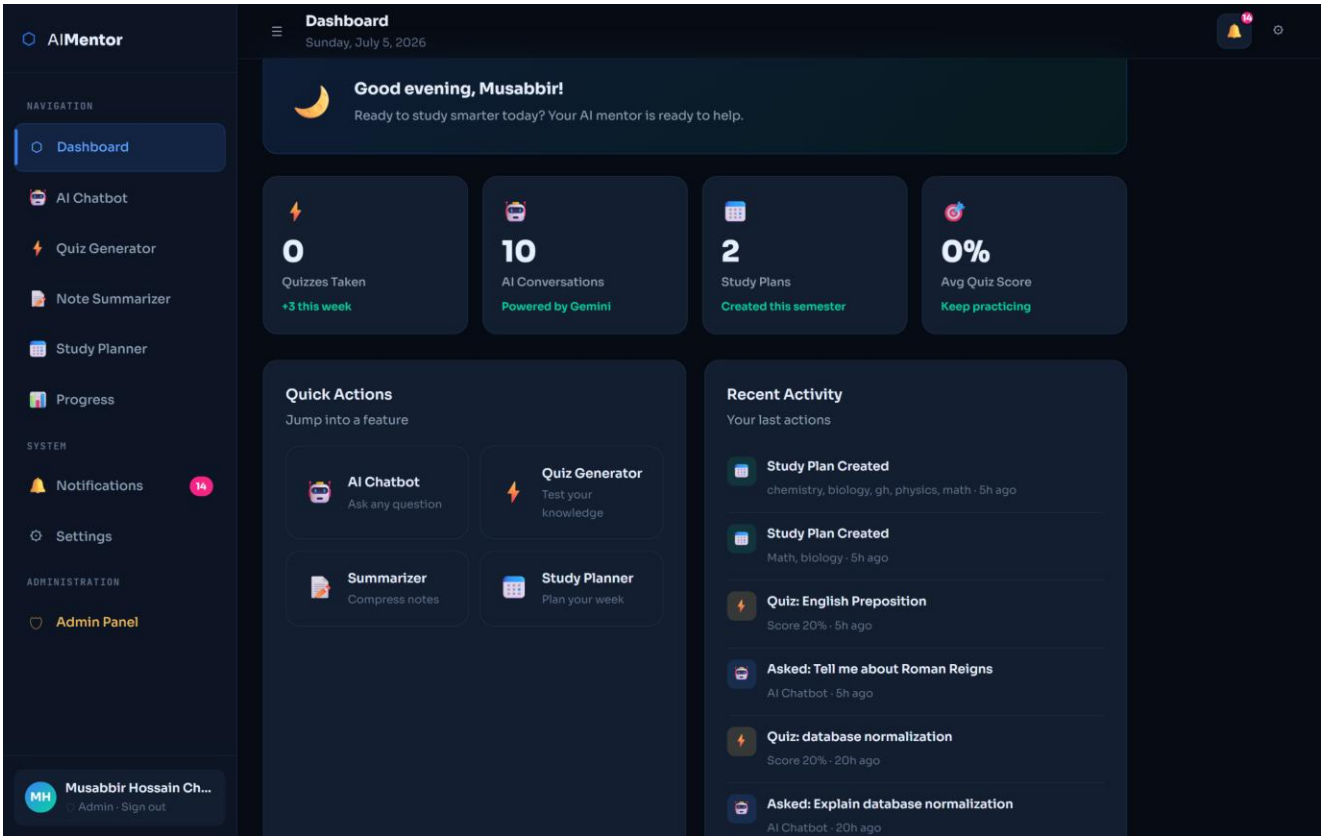
DESKTOP — 1440PX

Home



DESKTOP – 1440PX

Dashboard & Login



DESKTOP — 1440PX

Register & Settings

AI Mentor

Create account

Join AI Academic Mentor — It's free

FULL NAME

Your full name

EMAIL ADDRESS

you@nubtk.edu

STUDENT ID

DEPARTMENT

T123012XXXX

CSE

PASSWORD

Min 6 characters

CONFIRM PASSWORD

Repeat password

☒ Registering as Administrator

ADMIN ACCESS CODE

Enter admin access code

Required to create an admin account. Contact your supervisor for this code.

Create Account

Already have an account? [Sign in](#) →

← Back to home

AI Mentor

NAVIGATION

[Dashboard](#)

[AI Chatbot](#)

[Quiz Generator](#)

[Note Summarizer](#)

[Study Planner](#)

[Progress](#)

SYSTEM

[Notifications](#) 14

[Settings](#)

ADMINISTRATION

[Admin Panel](#)

MH Musabbir Hossain Ch...
Admin · Sign out

Settings

Manage your account and preferences

[Profile](#)

[Data Management](#)

[About](#)

Profile Settings

Your account information

Name

Musabbir Hossain Chayon

Email

musa@gmail.com

Student ID

11230121168

Department

Not set

Role

admin

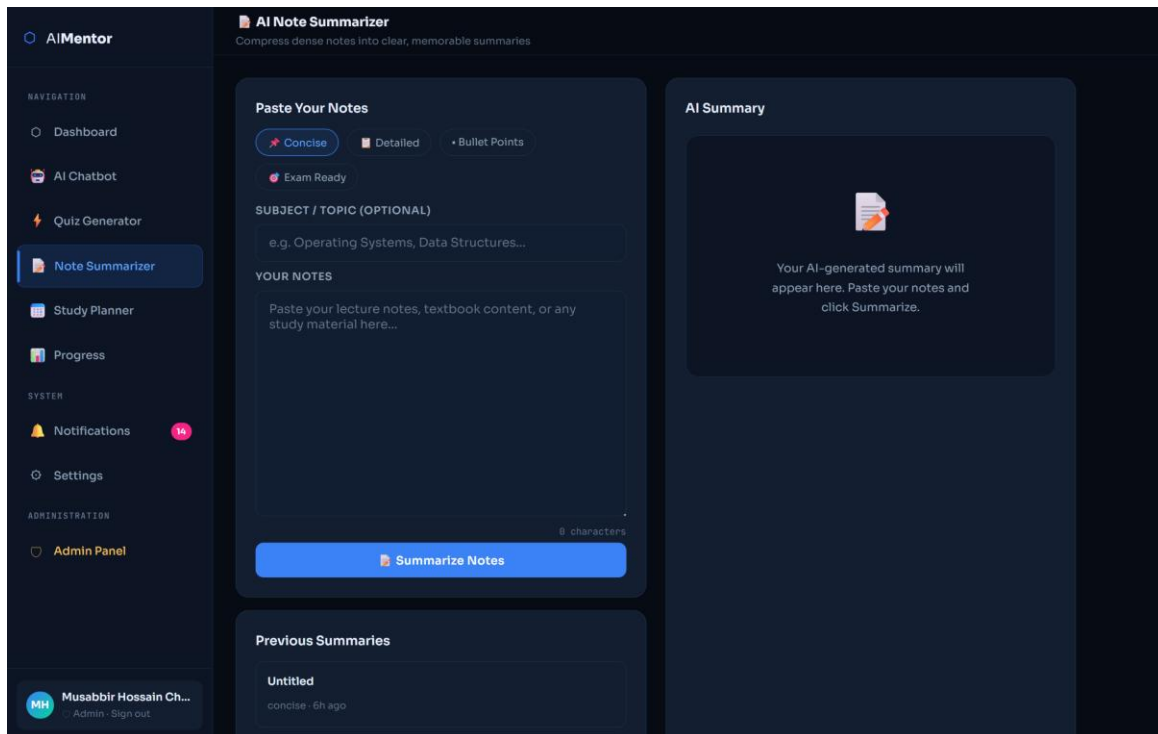
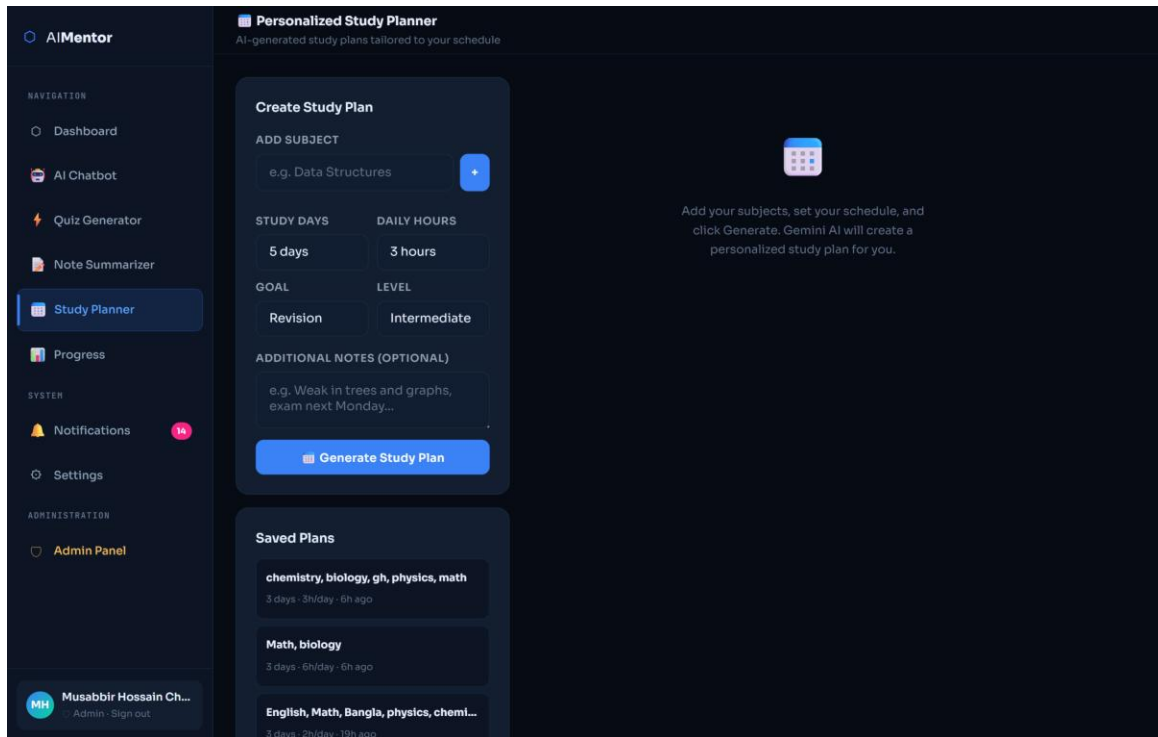
Joined

Unknown

Sign Out

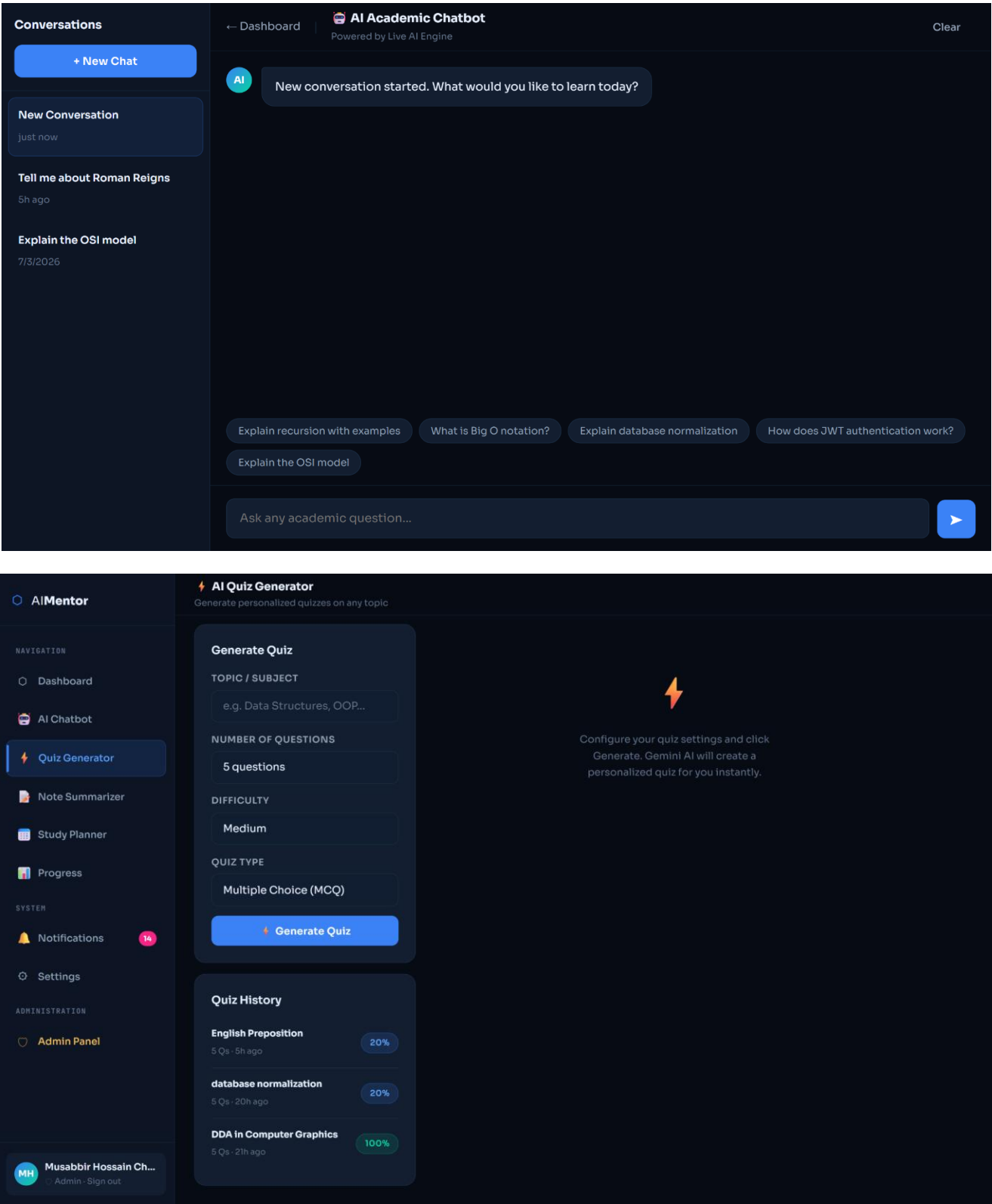
DESKTOP – 1440PX

Study Planner & Note Summarizer



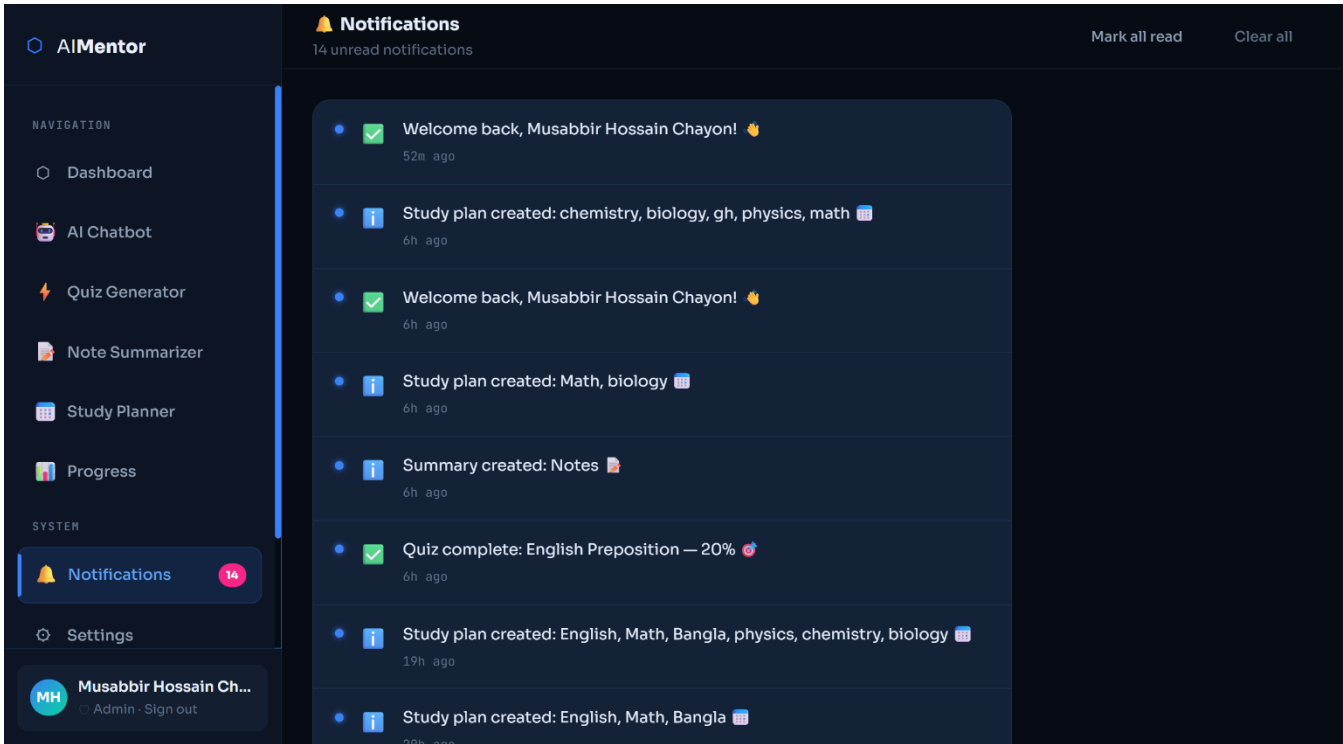
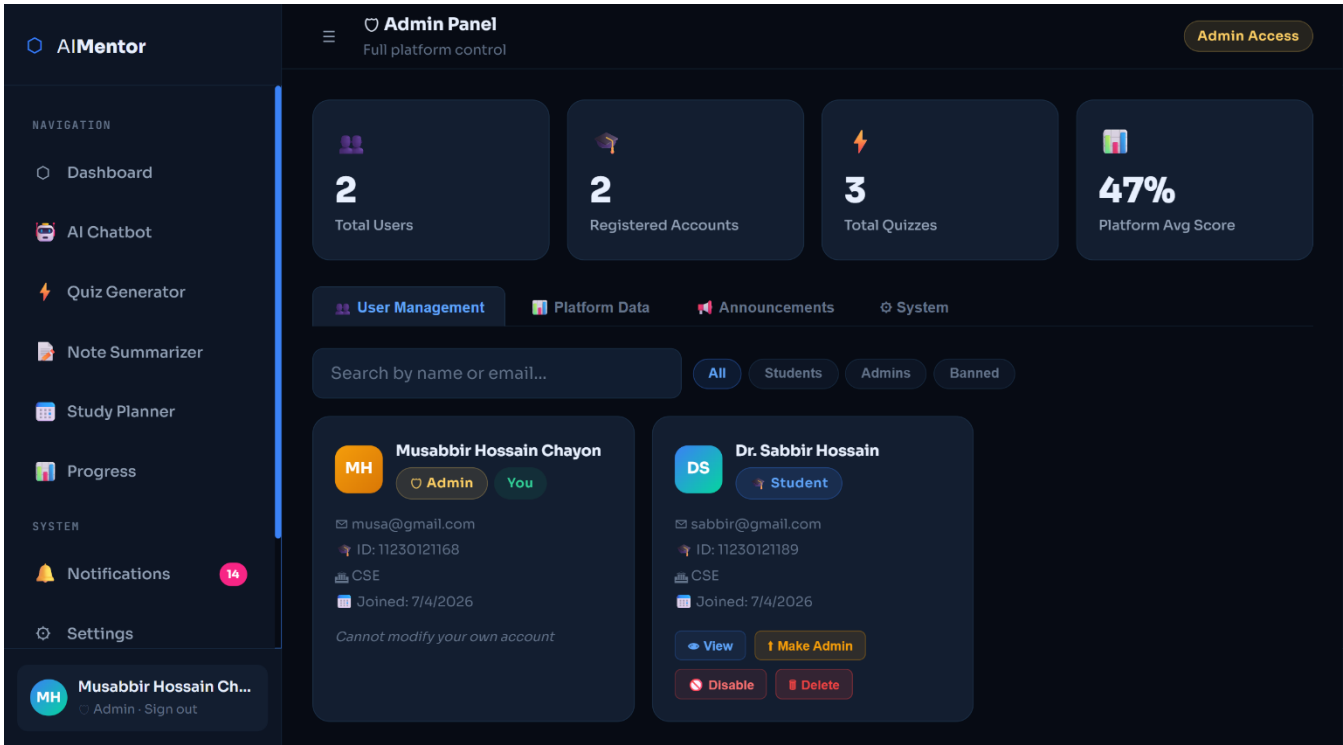
DESKTOP – 1440PX

AI Chatbot & Quiz Generator



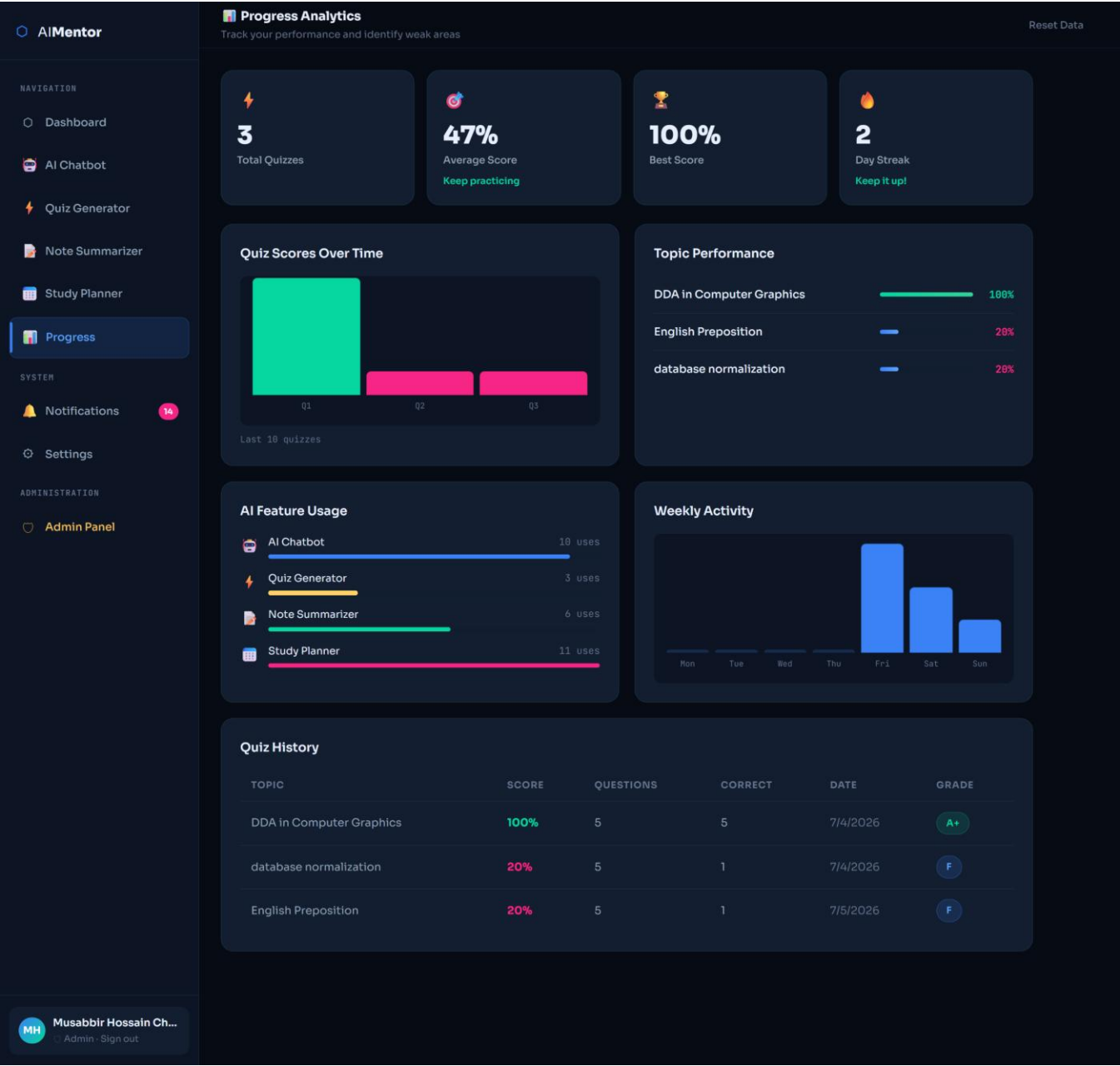
DESKTOP – 1440PX

Admin Panel & Notifications



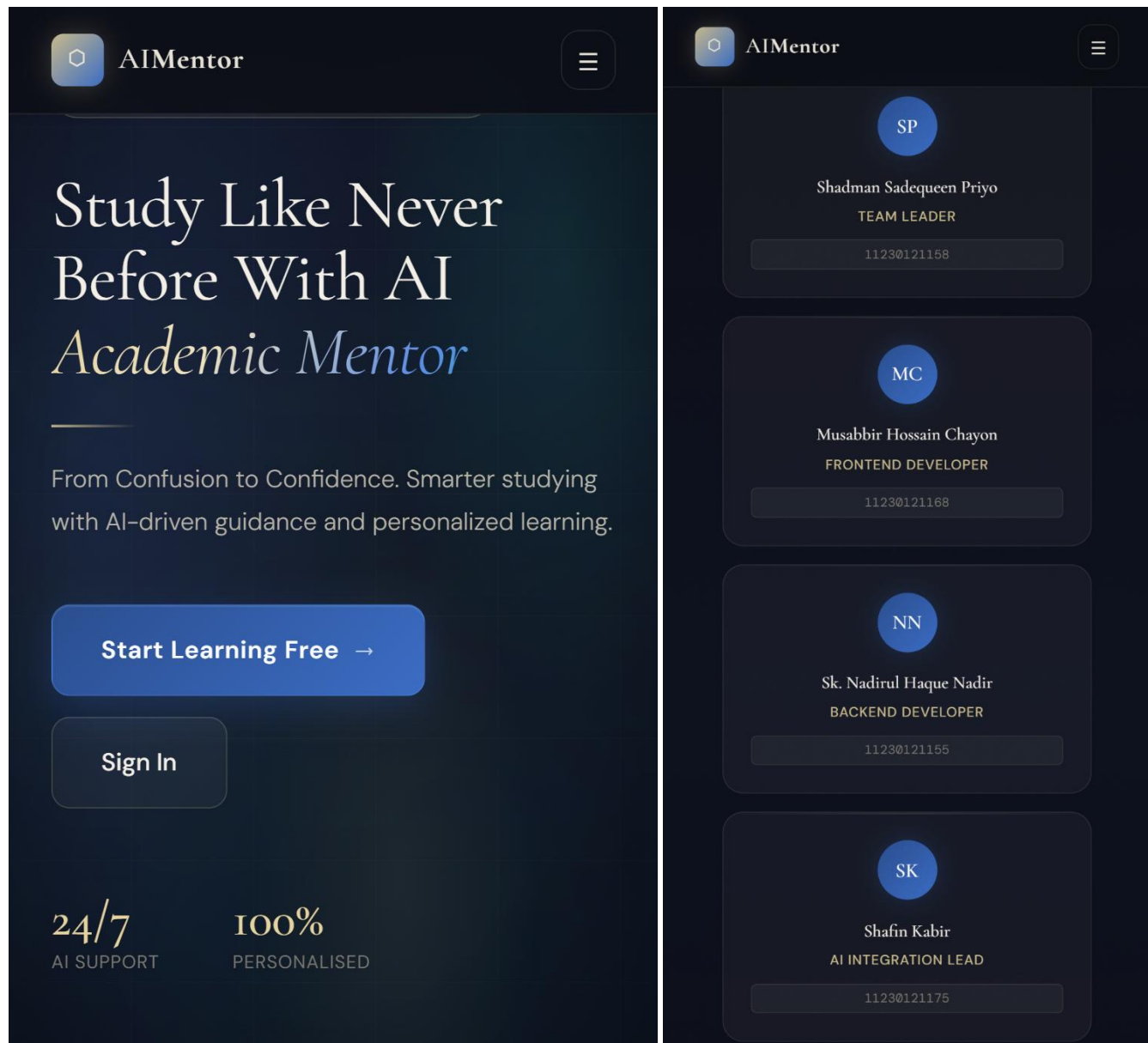
DESKTOP – 1440PX

Progress Analytics



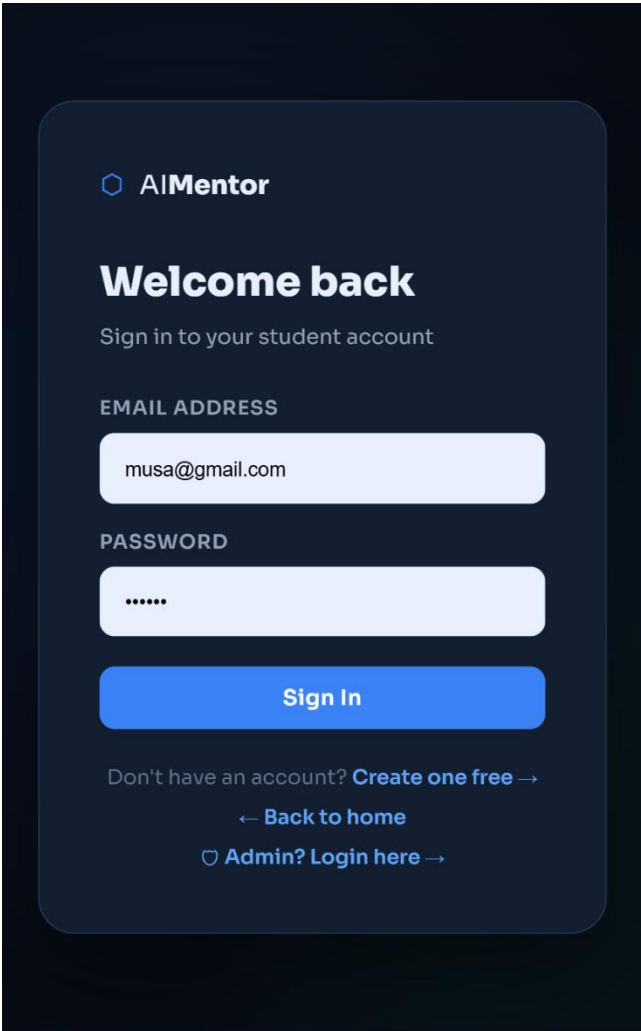
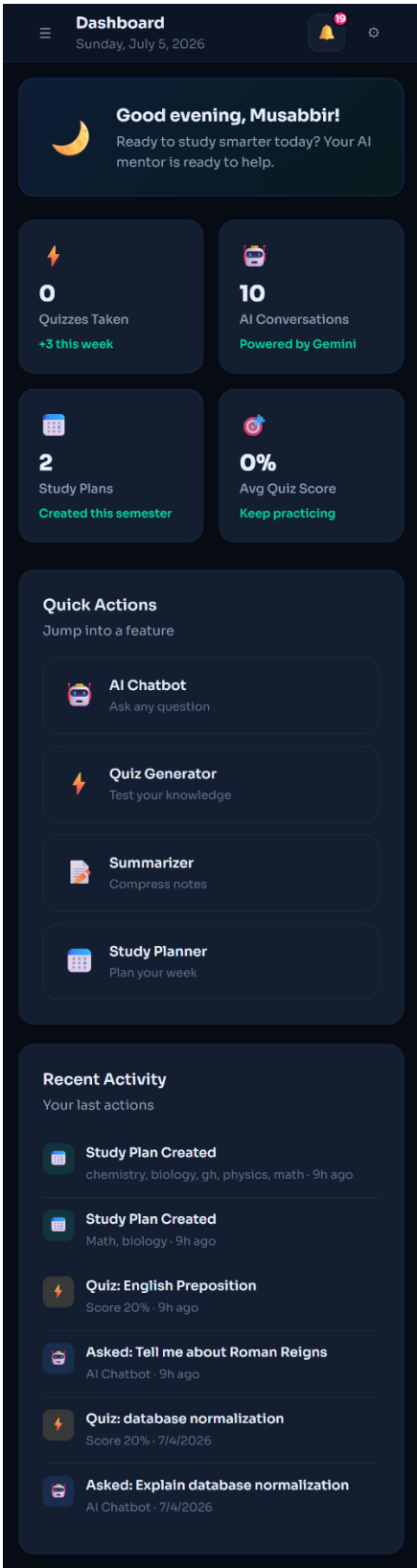
MOBILE – 425PX (RESPONSIVE VIEW)

Home



MOBILE – 425PX (RESPONSIVE VIEW)

Dashboard & Login



Register & Settings



Create account

Join AI Academic Mentor — it's free

FULL NAME

EMAIL ADDRESS

STUDENT ID

DEPARTMENT

PASSWORD

CONFIRM PASSWORD

☒ Registering as Administrator

ADMIN ACCESS CODE

Required to create an admin account. Contact your supervisor for this code.

Create Account

Already have an account? [Sign in](#) →

[← Back to home](#)

 Settings

Manage your account and preferences

 Profile

 Data Management

 About

Profile Settings

Your account information

Name

Musabbir Hossain Chayon

Email

musa@gmail.com

Student ID

11230121168

Department

Not set

Role

admin

Joined

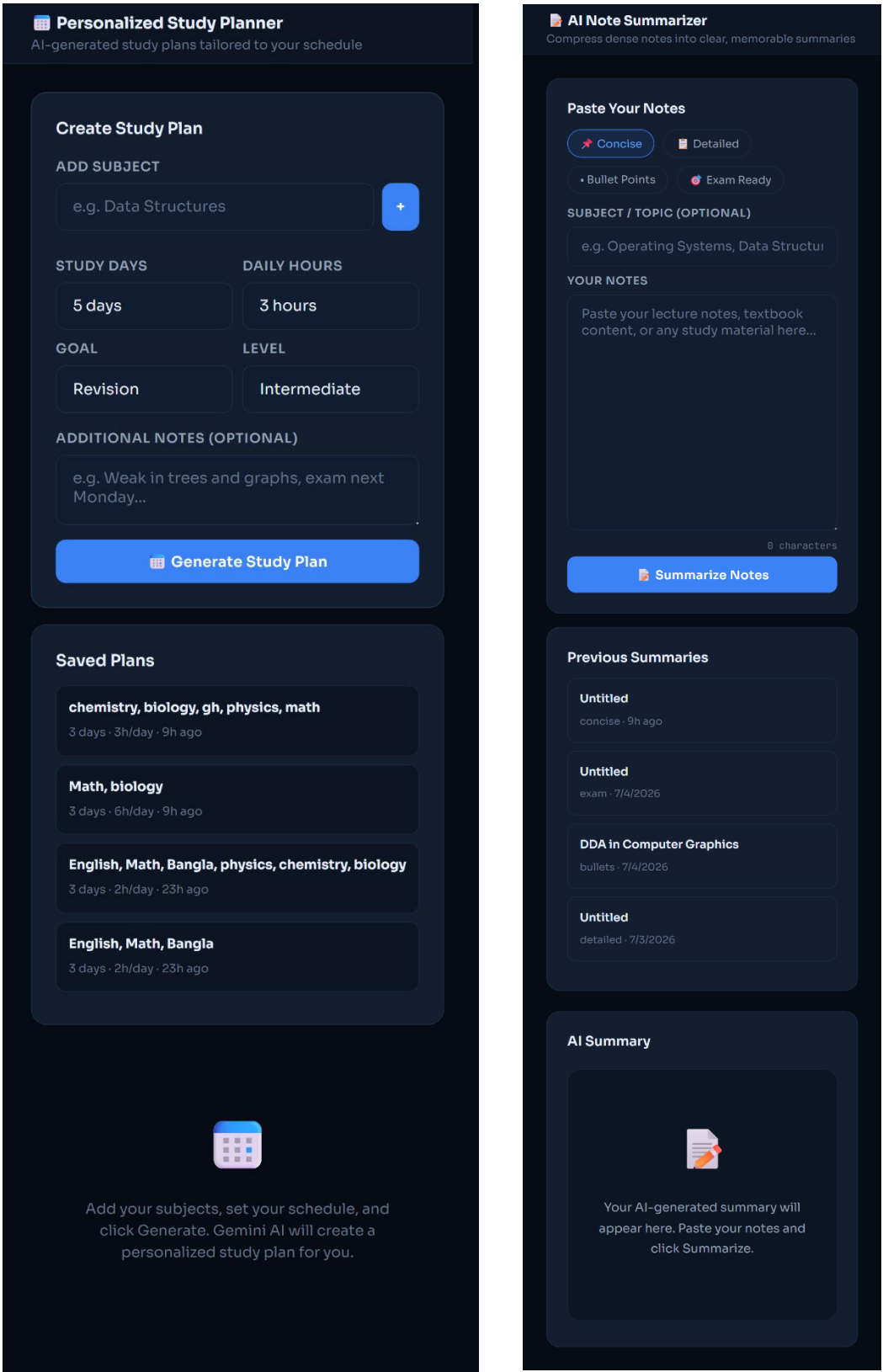
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Sign Out

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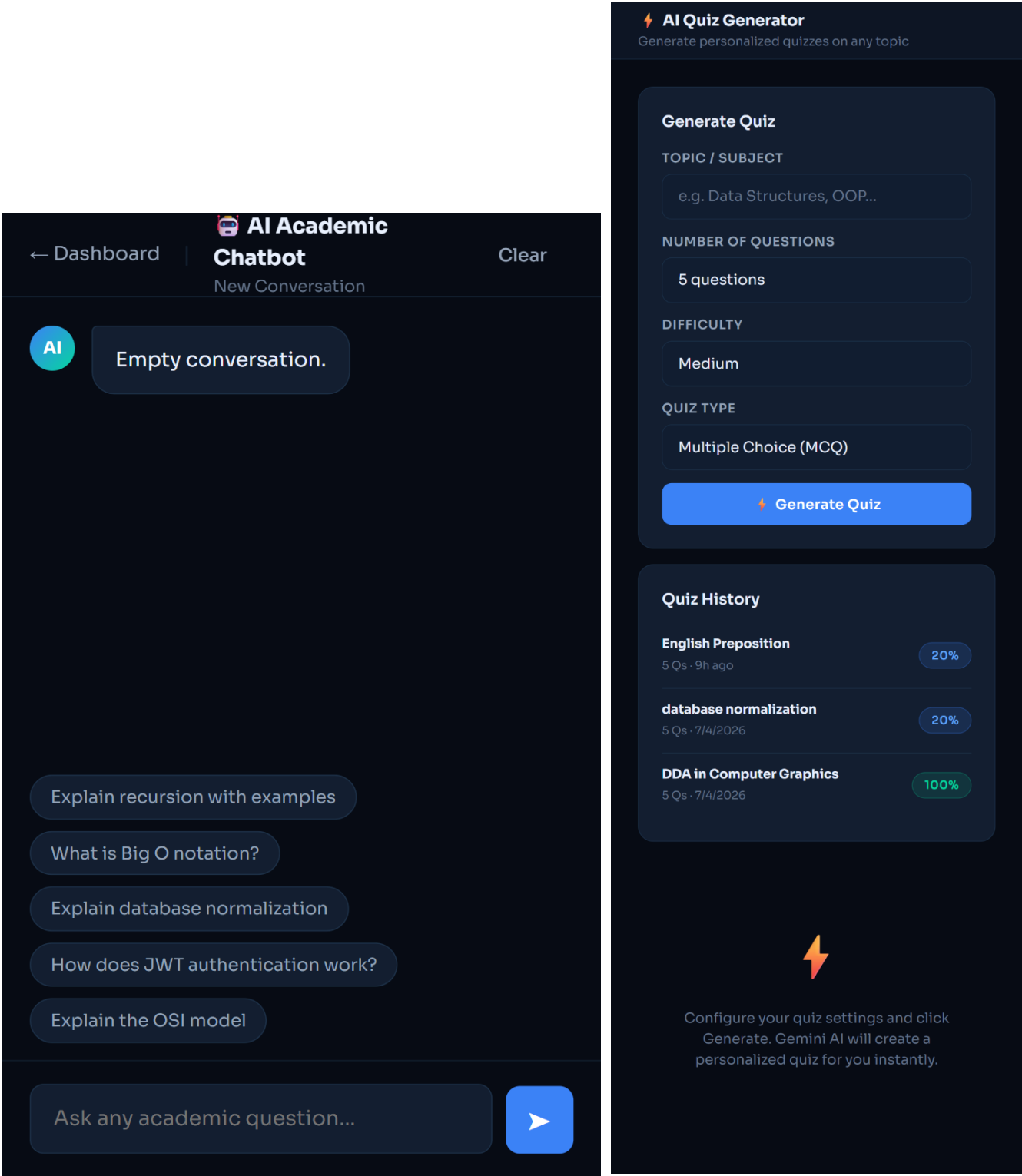
MOBILE – 425PX (RESPONSIVE VIEW)

Study Planner & Note Summarizer



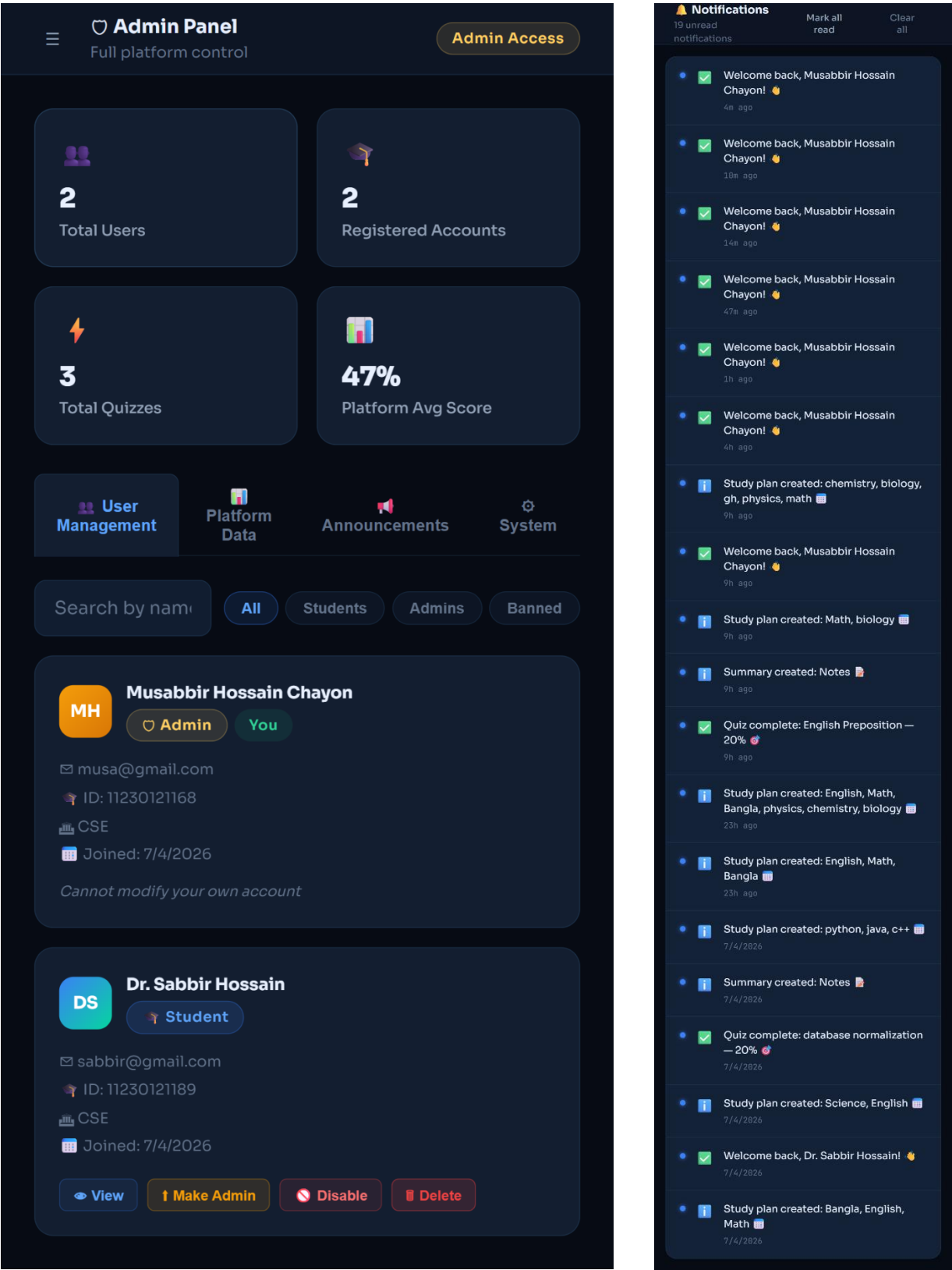
MOBILE – 425PX (RESPONSIVE VIEW)

AI Chatbot & Quiz Generator



MOBILE – 425PX (RESPONSIVE VIEW)

Admin Panel & Notifications



Progress Analytics

